

MIMU48XC-SD CARD

Integration Guide

Revision 0.5

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Revision History

Revision	Revision Date	Updates
0.1	10 Aug 2020	Initial version
0.2	12 Aug 2020	Included a command to store Precision IMU (with magnetometer) data.
0.3	13 Aug 2020	Data write sectors moved to (0 – d’9999) Reserved sectors (metadata) moved to (d’10000 – d’10038) Included a section “Important information about microSD card”
0.4	18 May 2021	Renamed the existing “Memory Reset” command to “Memory Write Pointer Reset”. Redefined “Memory Reset” command. New command for Precision IMU data along with calibrated magnetometer data stored in SD card where the option of selection of IMUs introduced. Change in memory structures Data Write Structures moved back to (d100- to end) Reserved sectors(metadata) moved to (0-d’38)
0.5	11 Nov 2022	Section “microSD Card Operation”: Described what to do when the session count reaches 31. Commands and corresponding description included in the Table 1 and Appendix 1 (i) Raw IMU Data (acc, gyro, mag) with SD card (ii) Get data for Raw IMU (acc, gyro, mag) session through USB only Removed commands and description from Table 1 and Appendix 1 (i) Output SD Card only through USB (ii) Get Data between sectors through USB only (iii) Get First & Last sectors of all session for SWDR data (iv) Get First & Last sectors of all session for Precision IMU data (v) Get First & Last sectors of all session for Raw IMU data

		Description updated for commands in the Table 1 and Appendix 1 (i) Writing Raw IMU data into SD card (ii) Reading Raw IMU data from SD card. Description updated for commands in Appendix 1 (i) Reset SWDR directory (ii) Reset Precision IMU directory (iii) Reset Raw IMU directory (iv) Get data for Precision IMU with calibrated Mag#4 data session through USB only Response updated for commands in Table 1 and Appendix 1(i) Get Total Sessions of SWDR (ii) Get Total Sessions of Precision IMU data (iii) Get Total Sessions of Raw IMU data. In response to these commands, 7bytes data packet comes first followed by the 4bytes ACK.
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Purpose & Scope

This describes the communication protocol for using MIMU48XC with microSD card. It describes how user can access & control the data and the processing at various stages, through an external application platform. This is a supplementary guide to *MIMU4844/MIMU48XC Integration Guide*. Hence you may have to refer that for some commands and data types.

microSD card operation:

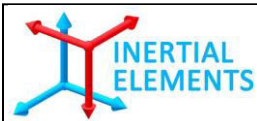
With SD card the MIMU48XC operates @ 200 Hz. It can collect SWDR, Precision IMU data and Raw IMU data* in microSD Card. For saving precision IMU data or raw data into microSD card one can give required command from Bluetooth or USB. However, for saving SWDR data one has to give the required start command from Bluetooth only. In order to read data stored in the microSD one has to give the required command only through USB. Reference scripts are available for writing and reading of data in SD card. Please read the part on using the script for further details.

SWDR data can be stored in SD card and simultaneously transmitted to the application platform in real-time. This scheme does not allow data to be lost even when the application platform is disconnected from the device for some time. All the SWDR data collected during disconnection are transmitted from the microSD card. This ensures robust handling of SWDR. The data collected in real-time can also be collected afterwards by accessing microSD card through USB.

The microSD card must be inserted only when the device is powered off. For proper functioning with the microSD card, it cannot be inserted when device is powered on.

Data can be stored in sessions. A session constitutes of the data stored between START command (specific to SWDR, Precision IMU or RAW IMU) and STOP (All output off) command. All the sessions' summaries corresponding to SWDR, Precision IMU and Raw IMU data are stored and maintain separately in the microSD card. Let's say one collects Precision IMU data 2 times followed by 3 SWDR sessions, followed by 4 Raw IMU session followed by 7 Precision IMU sessions, then total number of sessions for SWDR is 3, for Precision IMU is 9 and for Raw IMU data is 4. ~~One should remember that microSD card is capable of saving maximum of 31 sessions for each data type. Once 31 sessions are completed it again starts counting from session 1. All the data related to session 1 will be overwritten and so on.~~ User must keep track of number of sessions of each type, and must not allow it exceeding session count 31. The user can ensure this by resetting the type (SWDR or Precision IMU or RAW IMU) where the session count has touched 31. Resetting the particular type erases all the data corresponding to that type. The user may erase data of all the types (or the entire SD card) by giving the appropriate command.

The details of Start Address and End Address for each session corresponding to the Raw, Precision IMU & SWDR data are stored in the directory sectors. The last written address (write pointer), number of sessions for each kind of data along with the directory sectors are referred as *meta data* in this document.



MIMU48XC-microSDCard–Integration Guide

Doc: MIMU48XC -microSDCard-
Integration-Guide.pdf
Revision: 0.5
Release Date: 11th Nov, 2022

*Feature of collecting sensors' raw data from all the onboard IMU chipsets is available on the deliveries made after 1st August, 2020.

Important information about microSD card

The microSD card requirement:

- 1 GB or above
- Class 6 or above.

Storage capability of microSD card:

- Approx 44 hrs/ GB of Precision IMU (ax, ay, az, gx, gy, gz) @ 200Hz
- Approx 33 hrs/ GB of Precision IMU (ax, ay, az, gx, gy, gz, mx, my, mz) @ 200Hz
- Approx 2.5 hr/GB of Raw IMU data of all the 32 IMUs @ 200Hz
- SWDR data: Approx Two million steps' coordinates/GB.

Communication Protocol

The interfacing application platform (Computer, Phone, Tab etc) sends a particular command (data request) to MIMU48XC -SD CARD who serves the request by sending out required data to the application platform. Each command calls a particular state, which can send out variety of data depending upon command options.

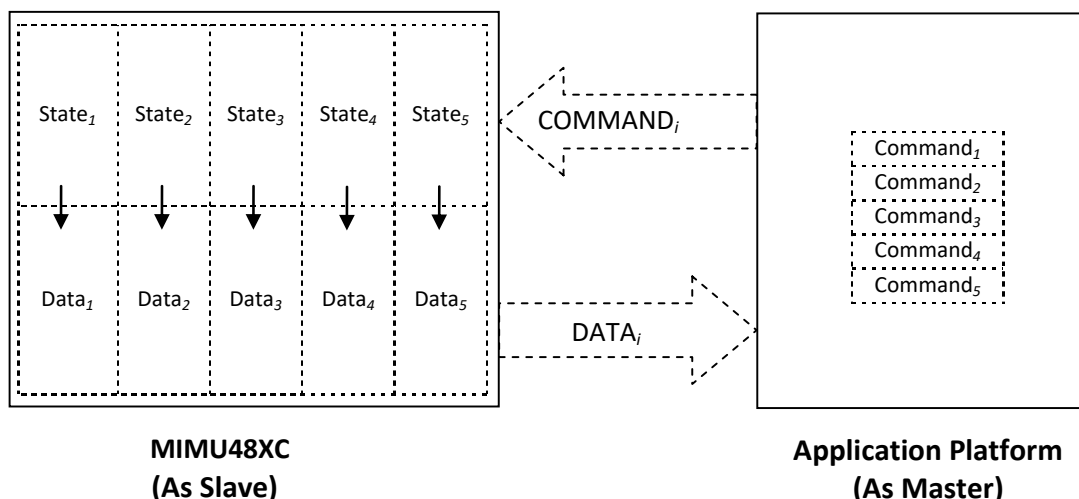


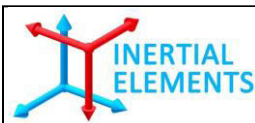
Figure 1 Data communication protocol between MIMU48XC -SD CARD and the application platform

Command format = [Command header, {payloads}, checksum]

Checksum is of two bytes (cs1, cs2). Last two bytes in any command are checksum.

Table 1: Summary of some useful commands

Sl. No.	Command	Description	Syntax (in decimal)
1	Stepwise Dead Reckoning with SD card (PDR)	MIMU48XC -SD CARD outputs displacement (dx, dy, dz) & change in orientation (dθ) w.r.t the previous detected step and entries of a 4x4 symmetric covariance matrix. The data is first written in the microSD card and then read and sent to the application platform. In this way no data is lost even if there is no transmission due to disconnection.	[69 0 69] cs1 = 0 cs2 = 69
2	Precision IMU Data with SD card	MIMU48XC saves a'x, a'y, a'z, g'x, g'y, g'z after calibration compensation and sensor fusion in microSD card.	[72 p11 cs1 cs2] p11 is the Output Rate Divider*
3	Precision IMU Data in navigational reference frame	MIMU48XC saves a'x, a'y, a'z, g'x, g'y, g'z in navigational (earth's)	[70 p11 cs1 cs2] p11 is the Output Rate Divider*



MIMU48XC-microSDCard-Integration Guide

Doc: MIMU48XC -microSDCard-Integration-Guide.pdf
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	with SD card	reference frame after calibration compensation and sensor fusion in SD card.	
4	Precision IMU Data & calibrated MAG#4 data with SD card	MIMU48XC saves a'x, a'y, a'z, g'x, g'y, g'z, m'x(4), m'y(4), m'z(4) after calibration compensation and sensor fusion in microSD card.	[89 pl1 cs1 cs2] pl1 is the Output Rate Divider*
5	Output SD Card only through USB**	This command outputs all the data from the microSD card through USB	{73 pl1 pl2 cs1 cs2} pl1 & pl2 are select size*
6	Memory Write Pointer Reset	This command resets the write pointer to the initial write address in the SD card. After this command, MIMU48XC starts overwriting all the old data.	[80 01 0 81]
7	Reset SWDR Directory	This command resets the SWDR directory in SD card, i.e. all the meta data related to SWDR will be reset. Session number corresponding to SWDR will be reset to 0 and all the corresponding start address of each session will be reset to 0.	[88 02 0 90]
8	Reset Precision IMU Directory	This command resets the Precision IMU directory in SD card, i.e. all the meta data related to Precision IMU will be reset. Session number corresponding to Precision IMU will be reset to 0 and all the corresponding start address of each session will be reset to 0.	[88 03 0 91]
9	Reset Raw Directory	This command resets the Raw IMU Directory in SD card, i.e. all the meta data related to Raw IMU will be reset. Session number corresponding to Raw IMU will be 0 and all the corresponding start address of each session will be set to 0.	[88 04 0 92]
10	Memory Reset	To erase all the data from the SD card use “Memory Write Pointer Reset” along with “Reset SWDR Directory”, “Reset Precision IMU Directory” & “Reset Raw Directory” commands.	[80 01 0 81] [88 02 0 90] [88 03 0 91] [88 04 0 92] All the above four commands must be used to reset the memory.
11	Get Total Sessions of SWDR ^{\$}	This command outputs total number of session for SWDR data that are stored in SD Card.	[82 00 82]
12	Get Total Sessions of Precision IMU data ^{\$}	This command outputs total number of session for Precision IMU data (corresponding to command no. 2, 3 or 4) that are stored in SD Card.	[81 00 81]
13	Get Data between sectors through USB only**	This command outputs data from the selected sectors from SD card, through USB.	{83 pl1 pl2 pl3 pl4 pl5 pl6 pl7 pl8 pl9 cs1 cs2} pl1 & pl2 are select size* pl3 pl4 pl5 pl6 is start sector pl7 pl8 pl9 pl10 is end sector cs1, cs2 are checksum

14	Get data for SWDR session through USB only**	This command outputs SWDR data for a particular session, through USB.	[84 00 64 p11 cs1 cs2] p11 is the session number
15	Get data for Precision IMU session through USB only**	This command outputs Precision IMU data for a particular session, through USB.	[85 01 254 p11 cs1 cs2] p11 is the session number
16	Get data for Precision IMU with calibrated Mag#4 data session through USB only**	This command outputs Precision IMU data + calibrated Mag#4 for a particular session, through USB.	[85 01 246 p11 cs1 cs2] p11 is the session number
17	Get First & Last sectors of all session for SWDR data	This command outputs all the first and last sectors for all the session that are stored in the SD card for SWDR	[87 00 87]
18	Get First & Last sectors of all session for Precision IMU data	This command outputs all the first and last sectors for all the session that are stored in the SD card for Precision IMU data	[86 00 86]
19	Raw IMU Data with SD card	MIMU48XC outputs ax, ay, az, gx, gy, gz of all the 32 IMUs as chosen	[90 p11 p12 p13 p14p15 cs1 cs2] p11, p12, p13, p14 all together consists of 32bits. Each bit corresponds to particular IMU. Thus p11-p14 is used to select IMUs. (p11; p12; p13 & p14 must be set to 0xff.) p15 is the Output Rate Divider *. cs1, cs2: checksum
20	Get Total Sessions of Raw IMU data ^{\$}	This command outputs total number of session for Raw IMU data that are stored in SD Card	[91 00 91]
21	Get First & Last sectors of all session for Raw IMU data	This command outputs all the first and last sectors for all the session that are stored in the SD card for Raw IMU data	[92 00 92]
22	Get data for Raw IMU session through USB only**	This command outputs Raw IMU data for a particular session, through USB	[93 01 138 p11 cs1 cs2] p11 is the session number
23	High precision IMU with calibrated magnetometer with selection of IMUs saved in SD card(Only valid for selection of any combinations of 1, 2, 4, 8, 16 and 32 IMUs. Selecting any other combination results in invalid data.)	MIMU4844/48XC outputs g'x, g'y, g'z, a'x, a'y, a'z after calibration compensation and sensor fusion and calibrated magnetometer of the lowest IMU# from different combination of selection of IMUs	[62 p11 p12 p13 p14 p15 cs1 cs2] p11, p12, p13, p14 all together consists of 32bits. Each bit corresponds to particular IMU. Thus p11-p14 is used to select IMUs. p15 is the output mode*. cs1, cs2: checksum
24	Raw IMU Data(acc, gyro, mag) with SD card	MIMU48XC outputs ax, ay, az, gx, gy, gz, mx, my, mz of all the 32 IMUs as chosen	[95 p11 p12 p13 p14 p15 cs1 cs2] p11, p12, p13, p14 all together consists of 32bits. Each bit corresponds to particular IMU. Thus p11-p14 is used to select IMUs. (p11; p12; p13 & p14 must be set to 0xff.) p15 is the Output Rate Divider *. cs1, cs2: checksum
25	Get data for Raw IMU (acc,	This command outputs Raw IMU	[93 02 00 p11 cs1 cs2]

	gyro, mag) session through USB only**	data (acc, gyro, mag) for a particular session, through USB	p11 is the session number
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*Output Rate Divider: Its 1 byte value (p11 for Precision IMU data and p15 for Raw IMU data) ranging from 0x0 to 0x0f. It manipulates the rate of output of the particular state asked in this command. If sensors' data sampling rate is 'f', then the output data rate of this particular state will be $f / (2^{(p11-1)})$. The largest value of the Output Rate Divider can be 15. Thus if p11 (or p15[3:0]) = 1 (smallest value), then the particular state will be stored @ 200 Hz in SD card and so on. ($f = 200\text{Hz}$)

Output Rate Divider	Data Transmission Rate (Hz)
1	200
2	100
3	50
4	25
5	12.5
6	6.25
7	3.25

***Select Size:** This gives the number of bytes one wants as output from a given sector. For SWDR only 64 bytes are written on each sector i.e. one packet on each sector. For Precision IMU data 510 bytes are written on each sector i.e. $510/34 = 15$ data packets on each sector. For Raw IMU data it saves 394 bytes if all the 32 IMUs are chosen.

** There will be no checksum and output header. This is due to huge size of the payload the payload is directly output instead of regular command sequence of storing it in state variables.

\$ In response to these commands, 7 bytes data packet comes first followed by the 4 bytes ACK.

Please keep in mind to select size properly otherwise you may receive junk data.

Output packet format: Output data packet is of the following format: [**Output header, payload, checksum**]

Each data packet contains

- 4 bytes output header which consists of packet header (1 byte), packet number (2 byte) and payload size (1 byte). There are commands which generate (i) no output packet (only ACK), (ii) only one output packet and (iii) multiple output packets. For (ii), packet number will always be 00 01. For (iii), packet number will indicate the sequence number of that particular packet.

- payload – This is different from command to command
- 2 bytes checksum.

Refer Appendix I for detailed description of Commands and States

Command ACK: For any of the command, if the SD card is not present then the ACK will be a normal i.e. [160 CH cs1 cs2] where CH is command header.

If SD card is present then the command ACK will be will be [160 CH+1 cs1 cs2] for all the commands.

Eg: If command is [69 00 69]

Command ACK will be (i) [160 70 00 230] if SD card is present and (ii) [160 69 00 229] if SD is not present.

All commands are followed by command ACK.

Appendix I

Command & its Output

1. Stepwise Dead Reckoning with SD card (PDR)

Command in Hex: [45 00 45]

Example Output: Same as SWDR output. Refer *MIMU4844/MIMU48XC Integration Guide* for data format. Here is an example output of 64 bytes through Bluetooth

AA 00 01 3A B6 77 68 0D 00 3F 28 E7 E7 43 FF FE FF 48 07 B0 FF ED 00 03 00 03 00 39 FF 60 08 10 00 0C FF FF 00 08 FF E4 FF 3A 07 D1 00 13 FF DC 00 0F 00 0A FF 31 07 8A 00 05 FA E4 00 08 0E 90

2. Precision IMU Data with SD card or Precision IMU Data in navigation reference frame with SD card

Command: [72 pl1 cs1 cs2]/[70 pl1 cs1 cs2]

pl1 is the output rate divider.

Example: The data stored in SD card is same as the Precision IMU command. Refer *MIMU4844/MIMU48XC Integration Guide* for data format. Here only the precision IMU data will be written on the SD card. For precision IMU w.r.t. navigational reference frame command, the output will same format except the acceleration value will be in navigation (earth's) reference frame.

Here is an example output in Hex

AA 17 6C 1C C3 E6 7A 98 40 74 22 11 40 7B 3D 0F 41 03 7D 75 3A 16 A9 04 BC 38 C7 5C 3B 44 67 1A 0B 3C
AA 17 6D 1C C3 EA 1A 5A 40 72 2A 66 40 74 6B 15 41 02 00 56 BB 81 90 AE BB C4 02 4C 3C 15 68 7D 0B F7
AA 17 6E 1C C3 EE 02 2C 40 6E 2F AD 40 72 CF 04 41 00 D0 13 3B CE D8 92 3C 7B 8D 07 BB 6F B7 7A 0D 76
AA 17 6F 1C C3 F1 EA 29 40 71 61 98 40 79 05 2B 41 02 66 13 39 DB E9 56 3C DD 6F 45 BB 88 B3 AC 0E 24
AA 17 70 1C C3 F5 D2 27 40 76 1F C3 40 79 BB 38 41 04 59 68 3B 3B F5 01 3C 54 2B 53 BB 4C 5A C9 0C EC
AA 17 71 1C C3 F9 BA 52 40 72 AA BA 40 7C 8D A5 41 03 A5 AA BB 8B 3A FE BB 25 09 28 3B CB 04 99 0E
DF
AA 17 72 1C C3 FD A2 29 40 73 B4 98 40 73 6C E6 41 02 5C 5B BB 4E 0E 26 BB F8 19 C5 3C 6C 9E F2 0E DE
AA 17 73 1C C4 01 8A 63 40 6C 6C 7B 40 76 29 56 41 01 49 FF 3A 85 78 8A 3C 05 64 A7 BC 1B 9A 3C 0C 0E

3. Precision IMU Data & calibrated Mag#4 data with SD card

Command: [89 pl1 cs1 cs2]

pl1 is the output rate divider

The stored data in SD card for this command is same as output format of the “High precision IMU with calibrated magnetometer # 4 data”. Refer *MIMU4844/MIMU48XC Integration Guide* for data format. Data consists of Output header followed by a’x, a’y, a’z, g’x, g’y, g’z (High precision IMU data in 4 byte single precision format) followed by m’x, m’y and m’z (calibrated magnetometer data corresponding to mag#4). The session corresponding to this command are maintained under Precision IMU directory. Reading the data from microSD card corresponding to this command is same as reading Precision IMU from microSD card.

~~4. Output SD Card (only for USB)~~

~~Command: [73 p11 p12 cs1 cs2]~~

~~p11 & p12 are select size~~

~~This command outputs all the data from SD card. Select the size as 64 bytes if the SD card only contains the SWDR data. But if the SD card contains both SWDR and Precision IMU data then it is safe to use 510 as select size. This command will only have a command header. It will not have any output header and checksum at the end.~~

~~Example Output: The output is all the information saved in SD card. The output includes the information (metadata) about each session saved in reserve sectors (d’1000000 to d’1000038) and the regular data of all the sessions.~~

5. Memory Write Pointer Reset

Command: [80 01 00 81]

This command resets the memory write pointer. This command sets the memory write pointer to first data sector (0x0).

It is an input command. There will be no output, except ACK, in response to this command.

It will output only ACK [160 81 00 241] if SD card is present, otherwise it will output [160 80 00 240] that means SD card is not present so the reset has not been done.

6. Reset SWDR directory

Command: [88 02 00 90]

In this command the SWDR directory sector will be reset. Session number corresponding to SWDR will be reset to 0 and all the corresponding start address of each session will be reset to 0. It is an input command. There will be no output, except ACK, in response to this command.

As with the above command, it will output only ACK [160 89 00 249] if SD card is present, otherwise it will output [160 88 00 248] that means SD card is not present so the reset has not been done.

7. Reset Precision IMU directory

Command: [88 03 00 91]

In this command the Precision IMU directory sector will be reset. This command resets the SWDR directory in SD card, i.e. all the meta data related to Precision IMU will be reset. Session number corresponding to SWDR will be reset to 0 and all the corresponding start address of each session will be reset to 0. It is an input command and there will be no output.

As with the above command, it will output only ACK [160 89 00 249] if SD card is present, otherwise it will output [160 88 00 248] that means SD card is not present so the reset has not been done.

8. Reset Raw directory

Command: [88 04 00 92]

In response to this command, the Raw IMU directory sector will be reset. This command resets the SWDR directory in SD card, i.e. all the meta data related to SWDR will be reset. Session number corresponding to Raw IMU will be reset to 0 and all the corresponding start address of each session will be reset to 0. It is an input command.

There will be no output in response to this command, except ACK. It will output ACK [160 89 00 249] if SD card is present, otherwise it will output [160 88 00 248] that means SD card is not present so the reset has not been done.

9. Memory Reset

Command(s): [80 01 0 81]
[88 02 0 90]
[88 03 0 91]
[88 04 0 92]

Sequence of these four commands results in memory reset. The command sequence resets the session number of all the sessions, irrespective of data types. It also resets the start & end addresses of each session corresponding to the SWDR, Precision IMU & Raw IMU data.

This command sequence resets the memory and free up the storage.

10. Get Total Sessions of SWDR

Command: [82 00 82]

This command outputs total number of session for SWDR data that are stored in SD Card.

In response to this command, 7 bytes data packet comes first followed by the 4 bytes ACK [A0 53 00 F3]

Example Output: If 4 session of SWDR has been completed then the output will be
AA 00 01 01 **04** cs1 cs2

11. Get Total Sessions of Precision IMU data

Command: [81 00 81]

This command outputs total number of session for Precision IMU data that are stored in SD Card.

In response to this command, 7 bytes data packet comes first followed by the 4 bytes ACK [A0 52 00 F2]

Example Output: If 3 sessions of Precision IMU has been completed then the output will be
AA 00 01 01 **03** cs1 cs2

12. ~~Get Data between sectors (for USB only)~~

Command:[83 pl1 pl2 pl3 pl4 pl5 pl6 pl7 pl8 pl9 cs1 cs2]

———— pl1&pl2 are select size*

pl3 pl4 pl5 pl6 is start sector

pl7 pl8 pl9 pl10 is end sector

cs1, cs2 are checksum

As in the case with Output SD card command all the data between the selected sectors from the SD card will be output. This command will not give any output header. Only output followed CH.

```
AA 17 6C 1C C3 E6 7A 98 40 74 22 11 40 7B 3D 0F 41 03 7D 75 3A 16 A9 04 BC 38 C7 5C 3B 44 67 1A 0B 3C
AA 17 6D 1C C3 EA 1A 5A 40 72 2A 66 40 74 6B 15 41 02 00 56 BB 81 90 AE BB C4 02 4C 3C 15 68 7D 0B F7
AA 17 6E 1C C3 EE 02 2C 40 6E 2F AD 40 72 CF 04 41 00 D0 13 3B CE D8 92 3C 7B 8D 07 BB 6F B7 7A 0D 76
AA 17 6F 1C C3 F1 EA 29 40 71 61 98 40 79 05 2B 41 02 66 13 39 DB E9 56 3C DD 6F 45 BB 88 B3 AC 0E 24
AA 17 70 1C C3 F5 D2 27 40 76 1F C3 40 79 BB 38 41 04 59 68 3B 3B F5 01 3C 54 2B 53 BB 4C 5A C9 0C EC
AA 17 71 1C C3 F9 BA 52 40 72 AA BA 40 7C 8D A5 41 03 A5 AA BB 8B 3A FE BB 25 09 28 3B CB 04 99 0E
DE
AA 17 72 1C C3 FD A2 29 40 73 B4 98 40 73 6C E6 41 02 5C 5B BB 4E 0E 26 BB F8 19 C5 3C 6C 9E F2 0E DE
AA 17 73 1C C4 01 8A 63 40 6C 6C 7B 40 76 29 56 41 01 49 FF 3A 85 78 8A 3C 05 64 A7 BC 1B 9A 3C 0C 0E
```


13. ~~Get First & Last sectors of all session for SWDR data/Precision IMU data/Raw IMU data~~

Command: ~~[87 00 87] / [86 00 86] / [92 00 92]~~

~~This command outputs all the first and last sectors for all the sessions that are stored in the SD card for SWDR or Precision IMU data or Raw IMU data.~~

~~Data format of the payload: Single byte (unsigned) session number followed by corresponding 4 bytes start address (unsigned integer) followed by corresponding 4 bytes end address (unsigned integer) and so on, upto the total number of session for that particular data type.~~

~~Let's say, for SWDR there are 2 sessions starting from 101 to 109 and 110 to 125 so the output will be~~

~~AA 00 01 16 01 00 00 65 00 00 00 00 6D 02 00 00 00 6E 00 00 00 7D cs1 cs2.~~

14. Get data for Precision IMU session for USB only

Command: [85 01 254 p11 cs1 cs2] (for Precision IMU session data) / [85 01 246 p11 cs1 cs2] (for Precision IMU + calibrated Mag #4 session data)

Here p11 is the session number for which data is being read. Please remember that for Precision IMU data and Precision IMU + calibrated Mag#4 data maintain same session directory. The output to this command is same as Precision IMU command & High precision IMU with calibrated magnetometer # 4 data respectively based on what data is stored for that particular session. Refer *MIMU4844/MIMU48XC Integration Guide* for data format.

15. Get Total Sessions of Raw IMU data

Command: [91 00 91]

This command outputs total number of session for Raw IMU data that are stored in SD Card.

In response to this command, 7 bytes data packet comes first followed by the 4 bytes ACK [A0 5C 00 FC]

Example Output: If 3 session of Raw IMU has been completed then the output will be
AA 00 01 01 **03** cs1 cs2

16. Raw IMU data with SD card

Command: [90 p11 p12 p13 p14p15 cs1 cs2]:

Raw IMU data (ax, ay, az, gx, gy, gz) from all the 32 IMUs are collected and stored in the SD card @ 200 HZ. pl1, pl2, pl3, pl4 all together consist of 32bits. Each bit corresponds to that particular IMU. Thus pl1 to pl4 are used for selecting the IMUs. The fields pl, pl2, pl3 & pl4 must always be 0xff, when saving data in microSD card. Data stored has the same format as “Output inertial data” command. Therefore the command is [90 255 255 255 255 pl5 cs1 cs2]. Refer *MIMU4844/MIMU48XC Integration Guide* for data format.

17. Get data for Raw IMU session for USB only

Command: [93 01 138 pl1 cs1 cs2]

Here pl1 is the session number for which data is being read.

This command will result in data packets of size 394 bytes each. Each data packet will have 12 bytes (2 bytes each for ax, ay, az, gx, gy and gz) each for all the 32 IMUs, 4 bytes of timestamp, 4 bytes of header and 2 bytes of checksum.

Data output is same as “Output Inertial Data”. Refer *MIMU4844/MIMU48XC Integration Guide* for data format.

18. Precision IMU Data & calibrated MAG#4 data with SD card with option to select different combination of 1, 2, 4, 8, 16 and 32 IMUs.

Command : [62 pl1 pl2 pl3 pl4 pl5 cs1 cs2]

Here pl1, pl2, pl3 and pl4 constitute one byte each. These are used for selecting IMUs. Consider the pl1, pl2, pl3 and pl4 all together constitute 4 bytes i.e. 32 bits, where each bit corresponds to each of the 32 IMUs, so if any bit at particular location is =1 then that IMU is activated , or if it is 0 then that IMU is off.; cs1, cs2: checksum bytes

With this command, MIMU4844 can be used as a single high precision IMU. The modules are capable of giving calibration compensated and fused data from the IMU array. This means the module can be used as a single precision IMU which outputs 3-axis acceleration and 3-axis gyroscope data and calibrated 3 axis magnetometer data of the lowest IMU# selected. **Remember only 1,2,4,8,16 and 32 selection of IMUs are valid. Any other selection will give wrong output.** For details of the command you can also refer Command#71 in “*MIMU4844/MIMU48XC Integration Guide*”. To read the data corresponding to this command use Get data for Precision IMU session for USB only “[85 01 250 pl1 cs1 cs2]”. Output format is same as “Precision IMU Data & calibrated MAG#4 data with SD card”.

19. Raw IMU data (acc, gyro, mag) with SD card

Command: [95 pl1 pl2 pl3 pl4pl5 cs1 cs2]:

Raw IMU data (ax, ay, az, gx, gy, gz, temp, mx, my, mz) from all the 32 IMUs are collected and stored in the SD card @ 200 HZ. pl1, pl2, pl3, pl4 all together consist of 32bits. Each bit corresponds to that particular IMU. Thus pl1 to pl4 are used for selecting the IMUs. The fields pl, pl2, pl3 & pl4 must always be 0xff, when saving data in microSD card. Data stored has the same format as “Output osmium data” command. Therefore the command is [95 255 255 255 255 pl5 cs1 cs2]. Refer *MIMU4844/MIMU48XC Integration Guide* for data format.

20. Get data for Raw IMU (acc, gyro, mag) session for USB only

Command: [93 02 00 pl1 cs1 cs2]

Here pl1 is the session number for which data is being read.

This command will result in data packets of size 854 bytes each. Each data packet will have 4bytes timestamp, 12 bytes acc and gyro (2 bytes each for ax, ay, az, gx, gy and gz) each for all the 32 IMUs, 4 bytes of timestamp, 2 bytes each for all the 32 IMUs temperatures, 12 bytes mag (4bytes float each for mx, my and mz) each for all the 32 IMUs, 12 bytes (4 bytes float each) for fused mx, my and mz, 4 bytes of header and 2 bytes of checksum.

Data output is same as “Output Osmium Data”. Refer *MIMU4844/MIMU48XC Integration Guide* for data format.

Appendix II

How to use the scripts to store and read SWDR data, Precision IMU and Raw IMU data in SD Card (Offline Mode):

Scripts will be provided for reference and one can build one's own script by referring those and the commands.

Steps are similar for both the Precision IMU and Raw IMU.

Steps for writing data into SD card:

- Step 1: Open “use_as_raw_imu_store_sd.m”/ use_as_precision_imu_store_sd.m /use_as_swdr_store_sd.m based on what data one needs to collect.
- Step 2: Change the COM Port -Based on if one is using USB or Bluetooth to write data into SD card. For SWDR data, one can send command through Bluetooth only.
- Step 3: Run the script.
- Step 4: Data will be written in the SD card until one presses STOP button (a pop-up on running script).
- Step 5: Proceed to read store data.

Steps to read data from the SD card:

Use only USB to read data from SD card.

- Step1: Open“use_as_raw_imu_read_sd.m” / “use_as_precision_imu_read_sd.m” /“use_as_swdr_imu_read_sd.m” / based on what data one collected before.
- Step 2: Change the COM Port accordingly.
- Step 3: Run script.
- Step 4: Press STOP when popped up.
- Step 5: Data are stored in .mat file
- Step 6: One can save the .mat file data in .txt format.

** For RAW IMU data it takes same time to read data as it takes to write data. So, if one collects data for 30 minutes one may have to wait for 30 minutes for the STOP pop-up, to read all the collected data from SD card.

** For Precision IMU data it reads 15 times faster than it takes to write. So, if one collects data for 30 minutes one may have to wait for 2 minutes for the STOP pop-up to read all the collected data from SD card.

** For SWDR data it reads ~at least 200 times faster (even more than that) than it takes to write. So if one collects data for 30 minutes one may have to wait for 9 second for the STOP pop-up to read all the collected data from SD card.

Appendix III

SD Write command from Smartphone through Bluetooth (Offline Mode)

One may install

https://play.google.com/store/apps/details?id=de.kai_morich.serial_bluetooth_terminal&hl=en_IN or any similar application for Android phone.

Send the write command for SWDR / Precision IMU or Raw Command from the application. One may disconnect the Bluetooth in between and connect again to send the STOP command. (0x22 0x00 0x22). For SWDR, one needs to ignore any data packet in between.

Appendix IV

Stepwise Dead Reckoning with SD card & Android Application Platform (Real time Mode)

As an example, we consider a specific case of using MIMU48XC with SD Card for stepwise dead reckoning with an (Android) application platform, communicating wirelessly.

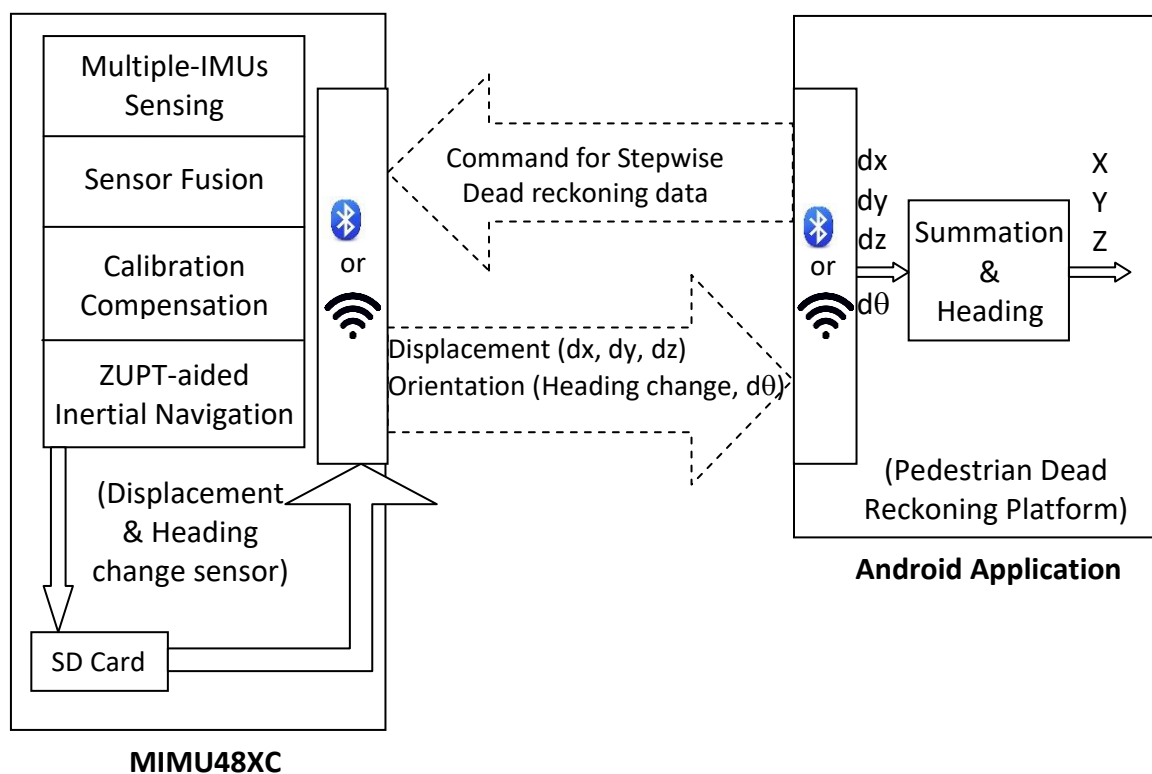


Figure 2 Interfacing of MIMU48XC with SD card and an application platform

